# Three Routine Intraoperative Variables Predict Vasoplegia at Separation from Cardiopulmonary Bypass

*No detectable loss of discrimination against a 38-parameter machine learning model in 704 adult cardiac surgery patients*

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KEY FINDINGS

**Three numbers already on every perfusion record predict vasoplegia at the moment of separation as well as a 38-parameter machine learning model.**

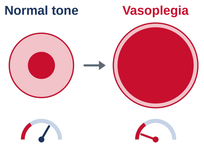
**0.858 vs 0.855**
AUROC of 3 routine variables vs the 38-parameter elastic net (difference +0.003, 95% CI −0.020 to 0.027)

**+0.08 to +0.09**
AUROC added by the minute-level flowsheet to the same algorithms on static data (Holm-adjusted p = 0.010)

**0.84 vs 0.57**
Among 438 patients not yet hypotensive at separation: three-variable model vs terminal MAP alone

**256 vs 3,209**
Patients needed (Riley) for a pre-specified 3- vs a 38-parameter model; this cohort has 704

## 1 Background: a late-recognized shock state




**Vasoplegia** is distributive shock after cardiopulmonary bypass (CPB). Vascular tone collapses and arterial pressure falls **while the heart itself pumps normally**.

Treatment escalates through vasopressors to rescue agents such as methylene blue, hydroxocobalamin, and angiotensin II.

- Reported incidence spans 2.5% to 66%, largely because definitions differ [1, 2].
- Associated with longer intensive care stays and higher mortality [3].
- Usually recognized only after vasopressor needs escalate, although earlier rescue therapy is associated with better survival [4].

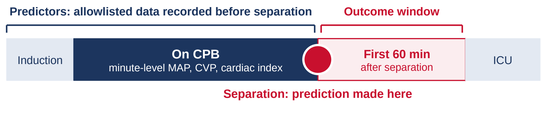
**Where vasoplegia risk tools operate**

| Before surgery | At separation from CPB | In the ICU, after surgery |
|---|---|---|
| clinical score, C = 0.68 [5] | this study | ML, AUROC 0.74 to 0.75 [6] |

*Different cohorts, definitions, and time points; context only, not a head-to-head comparison.*

**Aim** Early on-bypass hypotension was linked to vasoplegia with a single variable in 2009 [7], but no model has measured what the **minute-level intraoperative record** adds at the moment of separation, or how much model complexity a cohort with 45 events can support.

## 2 Study design and cohort



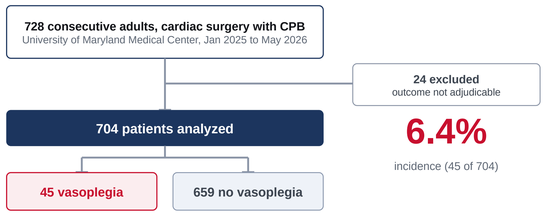

**Outcome, fixed before modeling (all three in the first hour after separation)**

**MAP < 65 mmHg**
averaged across the first hour

**CVP > 8 mmHg**
adequate filling, not hypovolemia

**Preserved cardiac function**
cardiac index > 2.2 L/min/m², or EF ≥ 55% without a PA catheter

**Pressure based by design:** no vasopressor-dose criterion, so the label is not conditioned on the treatment a deployed score would trigger. **Label check:** systemic vascular resistance index (SVRI) was lower in events (median 1,314 vs 1,784 dyn·s·cm$^{-5}$·m², p < 0.001; 338 patients with catheter data, 23 events). SVRI shares the defining pressures, so this confirms consistency only.




**Cohort characteristics by outcome**

| Characteristic | Vasoplegia (45) | None (659) | p |
|---|---|---|---|
| **Age, years** | 64 [55, 70] | 63 [55, 71] | 0.75 |
| **Male sex** | 32 (71%) | 451 (68%) | 0.87 |
| **Diabetes** | 12 (27%) | 175 (27%) | >0.99 |
| **CABG** | 16 (36%) | 356 (54%) | 0.020 |
| **Preop hemoglobin, g/dL** | 10.2 [8.9, 14.0] | 13.1 [11.1, 14.4] | 0.004 |
| **Preop creatinine, mg/dL** | 1.15 [0.91, 1.43] | 0.99 [0.80, 1.20] | 0.006 |
| **Preop INR** | 1.10 [1.00, 1.20] | 1.00 [1.00, 1.10] | 0.007 |
| **Aortic surgery** | 14 (31%) | 111 (17%) | 0.025 |
| **CPB duration, min** | **187 [146, 222]** | 116 [90, 156] | <0.001 |
| **Cross-clamp duration, min** | **122 [91, 159]** | 82 [59, 111] | <0.001 |
| **Mean MAP on bypass, mmHg** | **68.8 [65.2, 72.1]** | 74.8 [71.2, 78.3] | <0.001 |
| **Bypass minutes with MAP < 65** | **35% [24, 51]** | 22% [14, 31] | <0.001 |

*Median [IQR] with Mann-Whitney p, or n (%) with Fisher exact p.*

## 3 Predictors, with leakage control

**Static block** — 27 parameters
- Demographics and comorbidities
- Preoperative labs: creatinine, hemoglobin, INR, platelets, BUN, lactate
- Operative category
- Bypass and cross-clamp duration
- Cardioplegia and hemoconcentration volumes

**Minute-level block** — 11 parameters
- MAP, CVP, cardiac index summaries
- Fraction of bypass minutes with MAP < 65
- Mean MAP and CVP, final 15 min
- MAP slope, final 30 min
- Pulmonary artery catheter indicator

**38 candidate parameters**, all known at the moment of separation

**Excluded by allowlist:** anything recorded after separation, the outcome-defining first-hour averages, and intraoperative vasopressor or rescue-drug exposures. **Missing values:** training-fold median imputation inside every split (cardiac index 48% missing, preoperative lactate 69%).

## 4 Six models, one validation scheme

**Elastic net, full** — 38
penalized logistic; nested tuning; class-weighted

**Random forest, full** — 38
tree ensemble; captures interactions and thresholds

**Elastic net, static** — 27
same algorithm without the minute-level block

**Random forest, static** — 27
same algorithm without the minute-level block

**Clinical reference** — 7
L2 logistic: age, CPB and clamp time, Hb, creatinine, diabetes, hypertension

**Three-variable logistic** — 3
mean bypass MAP, CPB duration, hemoconcentration volume; no tuning

**Three-variable model:** the predictors the elastic net kept in all 10 folds with the largest standardized coefficients (one from the correlated pressure block), refit by unweighted maximum likelihood.

**Nested cross-validation, scored strictly out of fold**




Rotate the test fold 10 times → **pooled out-of-fold predictions for all 704 patients** → AUROC, AUPRC, calibration. The three-variable model used the same outer folds. Key comparisons repeated over **20 random re-partitions** with a fixed-hyperparameter elastic net (single-partition AUROC 0.848). Reporting follows the TRIPOD+AI statement for prediction models [8].

## 5 Statistics and robustness checks

**AUROC:** chance that a patient with vasoplegia gets a higher predicted risk than one without (0.5 = coin flip, 1.0 = perfect), with DeLong 95% CIs and paired tests [9] and Holm correction over 4 pre-specified contrasts.

**AUPRC:** area under the precision-recall curve. Its no-skill floor equals prevalence (0.064), so it rewards finding the rare events.

**Calibration:** do predicted risks match observed rates? Slope and observed:expected (O:E) of 1 are ideal; slope < 1 signals overfitting. 2,000 bootstrap resamples [10].

**Riley minimum n:** smallest cohort for which a model with p candidate parameters needs ≤ 10% shrinkage at its Cox–Snell R² [11]. EPV = events per parameter.

**Strata:** AUROC recomputed separately for patients with terminal MAP (last 15 min of bypass) of at least 65 mmHg and below 65 mmHg, separating prediction from detection.

**Pre-specified robustness checks** (pooled out-of-fold AUROC, DeLong 95% CI)

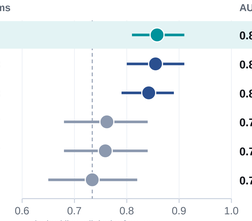
| | |
|---|---|
| Terminal MAP in place of mean bypass MAP | **0.849 (0.79 to 0.91)** |
| Both MAP summaries (four variables) | **0.874 (0.82 to 0.92)** |
| Six stable predictors | **0.869 (0.82 to 0.92)** |
| Three-variable model with class weighting | **0.855** |
| Outcome at MAP < 60 mmHg (16 events), three-variable | **0.876 (0.79 to 0.96); fold mean 0.89** |
| Outcome at MAP < 60 mmHg, full elastic net | **pooled 0.811; fold mean 0.88** |
| Full model without catheter indicator | **0.848** |
| Full model with ambiguous-timing totals | **0.864** |

*Full-model rows use the fixed-hyperparameter elastic net, except the MAP < 60 row (nested-tuned), whose lower pooled value reflects probability-scale shifts across folds at 16 events rather than worse ranking.*

## 6 The minute-level record adds 0.08 to 0.09 AUROC

**A Discrimination of all six models (pooled out-of-fold AUROC, DeLong 95% CI)**

| Model | Params | AUROC (95% CI) | AUPRC |
|---|---|---|---|
| Three-variable logistic | 3 | **0.858** (0.81 to 0.91) | 0.297 |
| **Elastic net · full** | 38 | **0.855** (0.80 to 0.91) | 0.323 |
| **Random forest · full** | 38 | **0.842** (0.79 to 0.89) | 0.290 |
| Elastic net · static | 27 | **0.762** (0.68 to 0.84) | 0.210 |
| Random forest · static | 27 | **0.759** (0.68 to 0.84) | 0.220 |
| Clinical reference | 7 | **0.734** (0.65 to 0.82) | 0.191 |

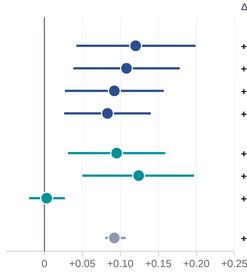



AUPRC no-skill floor = 0.064

**B Incremental value: paired difference in AUROC (95% CI)**

*EN = elastic net, RF = random forest*

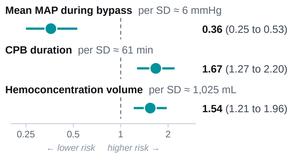
| | Δ AUROC (95% CI) | p |
|---|---|---|
| ***Confirmatory family, Holm-adjusted*** | | |
| EN full vs reference | **+0.120** (0.042 to 0.199) | **0.010** |
| RF full vs reference | **+0.108** (0.038 to 0.178) | **0.010** |
| EN full vs EN static | **+0.092** (0.027 to 0.157) | **0.010** |
| RF full vs RF static | **+0.083** (0.026 to 0.140) | **0.010** |
| ***Three-variable model (unadjusted p)*** | | |
| 3-variable vs EN static | **+0.095** (0.031 to 0.159) | 0.003 |
| 3-variable vs reference | **+0.124** (0.050 to 0.197) | <0.001 |
| 3-variable vs EN full | **+0.003** (−0.020 to 0.027) | 0.80 |
| ***20 re-partitions (percentile range)*** | | |
| EN (fixed) vs EN static | **+0.092** (0.080 to 0.107) | |




**What this means** With static data alone the models reach about 0.76; adding the minute-level flowsheet lifts them to about 0.86. Every confirmatory contrast survives Holm correction, and the three-variable model keeps the whole gain.

## 7 Three routine variables carry the signal

**C Odds ratio per SD, full-cohort fit**



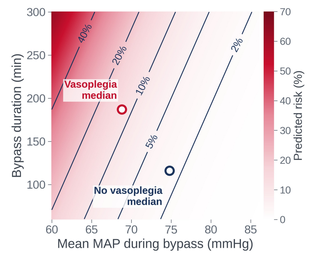

**The whole model fits on one line**

**lp = 8.193 − 0.1766 × MAP + 0.00843 × CPB min + 0.000423 × HC mL**
**risk = 1 / (1 + e$^{-lp}$)**

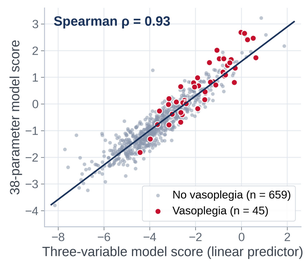
**Patient A:** MAP 76, CPB 110 min, HC 500 mL → **1.6%**
**Patient B:** MAP 68, CPB 190 min, HC 1,500 mL → **17.1%**

*HC = hemoconcentration volume. Hypothetical patients. Coefficient signs identical in all 10 training folds; 15 events per parameter.*

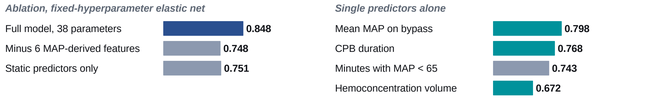
**D Predicted risk from the equation (hemoconcentration fixed at 1,000 mL)**




**E Out-of-fold scores of the two models, all 704 patients (linear predictors)**



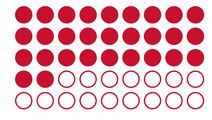

**F Where the signal lives (AUROC; bars start at 0.5 = chance)**




**What this means** Three variables explain 87% of the 38-parameter score, and removing MAP features erases its advantage over static data. The large model mostly re-derives the small one, consistent with evidence that machine learning rarely outperforms logistic regression on clinical data [12].

## 8 Prediction, not detection of hypotension

**The 45 vasoplegia cases, by MAP in the last 15 min of bypass**

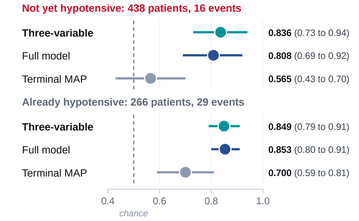



**Not yet hypotensive: 438 patients, 16 events**

| | |
|---|---|
| Three-variable | **0.836** (0.73 to 0.94) |
| Full model | **0.808** (0.69 to 0.92) |
| Terminal MAP | **0.565** (0.43 to 0.70) |

**Already hypotensive: 266 patients, 29 events**

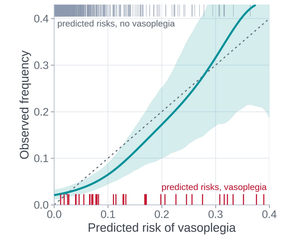
| | |
|---|---|
| Three-variable | **0.849** (0.79 to 0.91) |
| Full model | **0.853** (0.80 to 0.91) |
| Terminal MAP | **0.700** (0.59 to 0.81) |




**What this means** 29 of 45 events were already hypotensive before separation, so a model could score well just by reading the monitor [13]. Where pressure still looked normal, terminal MAP alone was near chance (0.57) while the three-variable model held at 0.84, because it reads accumulated pressure exposure and case burden rather than the current value.

## 9 Calibration and sample-size adequacy

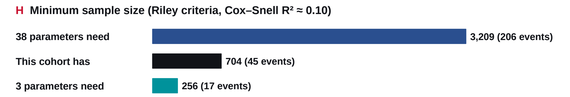
**G Calibration, three-variable model**




| | |
|---|---|
| **0.93** | **Calibration slope** 95% CI 0.72 to 1.20 (ideal 1) |
| **1.00** | **Observed:expected ratio** 95% CI 0.73 to 1.27 (ideal 1) |
| **0.00** | **Calibration intercept** 95% CI −0.40 to 0.32 (ideal 0) |
| **0.052** | **Brier score** vs 0.060 for a no-skill model |

*No recalibration step was applied.*

**H Minimum sample size (Riley criteria, Cox–Snell R² ≈ 0.10)**




**4.6-fold short** for 38 parameters (EPV 1.2) and **2.8-fold above** the need for 3 (EPV 15), if the three are treated as pre-specified, which selecting them from 38 candidates does not fully justify. Cox–Snell R² was 0.101 for the full model (cross-fitted, recalibrated) and 0.100 for the three-variable model.

## 10 Limitations and next steps

**Selection:** the three predictors were chosen once using all 704 patients before out-of-fold scoring, so 0.858 may be optimistic and the 3-parameter Riley figure assumes pre-specification; nested selection is planned.

**Inference:** p = 0.80 means no detectable difference, not formal equivalence; DeLong intervals on pooled predictions omit refitting and partition variance.

**Scope:** single-center retrospective study without external validation; 45 events overall, and only 16 among patients not yet hypotensive at separation.

**Data:** counts are provisional until a few screened records still under curation are resolved; cardiac index is missing in 48% and preoperative lactate in 69%.

**Label:** pressure based without a vasopressor criterion; at a 60 mmHg threshold only 16 events remain.

**Freeze model and preprocessing** → **Temporal validation in an expanded cohort** → **External multicenter validation** → **Prospective evaluation under intervention [14]**

**Expanded cohort:** a 3- or 5-year total window would add about 787 (≈ 50 events) or 1,781 (≈ 114 events) new patients; the final size will follow a validation-specific calculation [15].

**Conclusion** Three numbers on every perfusion record, mean bypass MAP, bypass duration, and hemoconcentration volume, predicted vasoplegia at separation from CPB as well as a 38-parameter machine learning model, with calibrated risks and within Riley sample-size criteria for a three-parameter model. The next step is validating the frozen model in new patients.

**References**

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